A Mutual Guide Framework for Training Hyperspectral Image Classifiers With Small Data

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***Abstract*— This article develops a general yet effective hyperspectral image (HSI) classification framework that is trained with small data. To this end, two identically structured but differently initialized classifiers, which are referred to as two base classifiers, are trained in an iterative manner. Each iteration consists of three steps, that is: 1) the two base classifiers that are trained separately on guide data; 2) unclassified data that are processed by the two trained base classifiers; and 3) the classification results with high confidence that are explored as new guide data. In the first iteration, the guide data comprising the original small training data are the same for the two base classifiers. From the second iteration, the guide data for the two base classifiers start becoming different. Specifically, in each iteration, the guide data for training one base classifier keep being augmented by high confidence classification results provided by the other base classifier. It is in such an iterative manner that the two classifiers continuously provide different new guide data for each other, and thus increasingly augment labeled data from the original small training set to a reasonably larger amount of samples in a HSI. We refer to such a training strategy as mutual guide. We develop a mutual guide implementation scheme by exploiting extreme learning machines (ELMs) as base classifiers. Extensive experiments on four public HSI datasets, i.e., Indian Pines (IP), Kennedy Space Center (KSC), University of Pavia (UP), and Salinas (SA), validate the classification effectiveness of our mutual guide framework with small training data.**

***Index Terms*— Hyperspectral image (HSI) classification, mutual guide, small training data.**

I. Introduction

RECENTLY, dramatic progress of hyperspectral imaging techniques with respect to various missions has been witnessed. With high resolution and abundant bands, remote sensing hyperspectral images (HSIs) precisely obtain continuous spectral information of different ground objects and have the advantage of exhibiting their subtle spectral difference [1]. HSIs play an important role in extensive practical scenarios such as modern agriculture [2], [3], environmental governance [4], [5], resource exploration [6], [7], and military defense [8], [9]. Hyperspectral image classification (HIC) is one fundamental task in HSI applications. HIC requires object classes to be precisely determined on the basis of the spectral characteristics of each pixel. Although the abundant spectral information provides great potential for information mining, the requirements for characterizing numerous spectral bands and extensive spatial information pose noticeable challenges to HIC [10]. Therefore, HIC has attracted a great surge of interests and has been comprehensively studied in the literature.

Initially, various shallow learning methods have been applied to HIC, including support vector machines (SVMs) [11], random forests (RFs) [12], and extreme learning machines (ELMs) [13]. Most traditional shallow learning methods just exploit spectral information rather than spatial information, and thus are not easy to achieve accurate results. Additionally, there are some spatial feature extraction methods such as image segmentation [14], [15], Gabor filter [16], [17], and superpixelwise principal component analysis (SuperPCA) [18], which address the insufficiency of the spatial information. Fang *et al*. [15] effectively utilized the spectral-spatial information of superpixels via multiple kernels. Jia *et al*. [16] developed a 3-D Gaussian-Gabor feature extraction and selection framework, which integrates spatial information with the spectral information for HIC. Jiang *et al*. [18] proposed SuperPCA that conducts principal component analysis (PCA) on each homogeneous region obtained by superpixel segmentation to incorporate spatial contextual information. SuperPCA circumvents the underlying indiscriminateness arising in PCA by regional processing rather than entire HSI manipulation. Shallow learning methods can obtain proper performance with the effective use of spectral-spatial information. However, their representational powers are limited due to straightforward architecture. When underfitted, they are deficient for big data classification.

In recent years, deep learning methods such as convolutional neural networks (CNNs) [19] and recurrent neural networks (RNNs) [20] have shown a great potential to portray the data internal information and yielded high performance in the

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HSIs [21]. Hu *et al*. [19] investigated supervised CNN for extracting deep features from HSIs. Hang *et al*. [20] proposed a cascaded RNN model, which processes the spectral features as sequential data, to explore the redundant and complementary information of HSIs. The classification performance of these models has not improved that of shallow learning methods significantly because of the neglect of the spatial information. 2-D convolutional neural networks (2DCNNs) [22] and 3-D convolutional neural networks (3DCNNs) [23], [24] were widely used to address the insufficient utilization of spatial information. Zhao and Du [22] developed a spectral-spatial framework that combines dimension reduction with 2DCNN for hyperspectral classification. Chen *et al*. [23] proposed a 3DCNN-based feature extraction model with combined regularization to extract effective spectral-spatial features of HSIs, thereby yielding higher classification performance. Contextual deep convolutional neural network (CDCNN) [25], which jointly exploits local spectral-spatial relationships of neighboring individual pixel vectors, was developed to improve HIC performance. Although these existing deep learning methods have achieved remarkable performance, there are some limitations. The powerful classification capabilities of the deep learning methods require training a large number of network parameters that demand numerous labeled data [26]. Additionally, most existing deep learning methods are extraordinarily sensitive to the quantity and quality of labeled data. In the situation of insufficient labeled data, they are prone to overfitting, which seriously deteriorates the classification performance.

Effectively using HSIs with rich spectral-spatial information requires comprehensive class labels. The labeling work is difficult and expensive, making the available high-quality labeled data limited. The lack of labeled data raises a situation that under certain circumstances HIC has to be done with small training data. Notwithstanding that the shallow learning methods and deep learning methods have achieved distinguished performance, most of them are not well appropriate to HIC with small training data. Classifying HSIs in the situation of small training data remains a long-standing challenge to be addressed.

To alleviate the above issue, some semi-supervised classification has been used to classify HSIs in an economical and effective manner, including active learning (AL) [27], co-training [28], and graph convolutional networks (GCNs) [29], [30]. Haut *et al*. [27] developed an AL model with Bayesian CNNs for spectral-spatial HIC. The AL intelligently selects and manually labels the most informative unlabeled data to the training set. Zhang *et al*. [28] extracted the original spectral features and the 2-D Gabor spatial features as two distinct views for co-training. GCNs [29] are emerging network architectures, which model relations between data in terms of graph representations [30]. Instead of only using labeled data for training, semi-supervised learning explores pseudo-labels predicted for unlabeled data and is thus more suitable for the situation of insufficient labeled data [31]. The involvement of pseudo-labels in training improves the classification performance of semi-supervised learning over those trained only with labeled data. However, it is hard to guarantee the correctness of the pseudo-labels and there may be wrong labels in the pseudo-labels. If the wrong labels are used as pseudo-labels for training, they become noisy labels and negatively affect the robustness of the classifiers. When noisy labels become a non-negligible proportion, the classification performance of semi-supervised learning deteriorates subject to the misleading training.

Considering the significance of HIC with small training data and the vulnerability of the semi-supervised learning methods subject to noisy labels, we develop a novel mutual guide framework. It trains two identically structured but differently initialized classifiers simultaneously in an iterative manner. Each classifier keeps providing new guide data with guide labels to the other. Though the iterations commence by using the same original small training data for training the two base classifiers, the guide data provided by the two base classifiers are different, because they are initialized differently and then gradually trained with different data. Therefore, the two base classifiers are trained quite independently such that each of them establishes a unique representational power for the same HSI. One reason that most semi-supervised learning methods are easy to generate noisy labeled data is that they tend to use pseudo-labels generated by one classifier to retrain the classifier itself, and self-confidence bias arises. In contrast, our mutual guide framework trains one classifier based on the guide data with guide labels provided by the other classifier. The independent training of the two base classifiers renders a more objective mutual confidence, and effectively neutralizes the one classifier self-confidence bias. Therefore, the mutual guide maintains a more robust training strategy with small data.

The effectiveness of the mutual guide framework can also be intuitively explained by considering the probably approximately correct (PAC) learning theory, in which the test error upper bound of a classifier with a certain VC dimension will become smaller when the small training data become larger. In our work, the training data amount increases with the guide data. In this scenario, the classifiers trained with guide data are expected to have better representational power than the original ones trained with the small data. Although the mutual guide framework is developed in the context of small data training, it is not limited to the small data condition and is applicable to normal training processes, in which the mutual guide behaves as a general approach to improving the representational power of classifiers.

The main contributions of the mutual guide framework are threefold.

1) We develop a mutual guide framework for HIC with small training data. In contrast to most state-of-the-art HIC methods that require big training data, the mutual guide achieves accurate HIC with small training data. The key to this effectiveness is that two base classifiers initially trained with the original small data keep independently providing high confidence classification results as guide labels to each other and in turn enhancing their training effectiveness separately. The mutual guide framework alternates between training and classification until a reasonable number of pixel samples

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in an HSI have been assigned confident class labels by the base classifiers. The augmented training data are sufficient for effective training. Therefore, the mutual guide framework is able to achieve accurate HIC in the condition of small training data.

2) The mutual guide framework is tolerant to intermediate noisy labels. The mutual guide framework addresses noisy labels, which possibly arise at the intermediate iterations, from two aspects. First, it manages to avoid the generation of noisy labels by mutual confidence and agreement between the two base classifiers. Second, even if small amounts of noisy labels arise at the intermediate iterations of mutual guide, the two classifier training strategy is validated to be immune to noisy labels. Therefore, the mutual guide framework properly addresses intermediate noisy labels, rendering a robust classification framework with small training data.
3) We develop a mutual guide implementation scheme based on the state-of-the-art SuperPCA as feature extractor and ELMs as base classifiers. One requirement for the base classifiers is that they should be initialized differently from each other. An ELM has the property that its input layer is randomly parameterized [13] such that it satisfies the requirement. The SuperPCA and ELMs form a baseline for implementing the mutual guide framework. We have released the baseline code,[1] which is open for substituting SuperPCA and ELMs by more effective feature extractors and base classifiers to validate the effectiveness of the mutual guide framework.

II. Mutual Guide With Small Training Data

This section describes the training and classification procedures of the mutual guide framework. It commences from introducing the functionalities of base classifiers, then describes the mutual guide training between two base classifiers, explains the classification procedures, and finalizes by analyzing the advantages of the mutual guide framework.

A. Base Classifier

Let $\boldsymbol{x} = [x_1, x_2, \ldots, x_M]^\top \in \mathbb{R}^M$ denote a sample to be classified and $\boldsymbol{y} = [y_1, y_2, \ldots, y_N]^\top \in \mathbb{R}^N$ denote the label indicator for $\boldsymbol{x}$, where $\top$ denotes the transpose operation and M and N are the dimensions of the sample and label indicator, respectively. For example, if $\boldsymbol{x}$ is a pixel sample in an original HSI, M is the number of the spectrum, and if $\boldsymbol{x}$ is a feature vector depicting a pixel sample in an HSI, M is the feature dimension. $\boldsymbol{y}$ can indicate the class of $\boldsymbol{x}$ in various ways. One widely adopted way is that if y_n is the biggest among all elements in $\boldsymbol{y}$, $\boldsymbol{x}$ belongs to the nth class. In this case, N is the total number of classes.

Let $C(\boldsymbol{x}; \boldsymbol{\alpha}, \boldsymbol{\beta})$ denote a classifier, where $\boldsymbol{\alpha}$ is the randomly initialized parameter and $\boldsymbol{\beta}$ is the model parameter to be trained. An ideal classifier $C(\boldsymbol{x}; \boldsymbol{\alpha}, \boldsymbol{\beta})$ follows the relation below:

$$\boldsymbol{y} = C(\boldsymbol{x}; \boldsymbol{\alpha}, \boldsymbol{\beta}). \tag{1}$$

[1] https://github.com/TaiXiaoxiao/MutualGuide

Let $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_P\}$ denote the whole dataset of P samples. It consists of the labeled data $\boldsymbol{X}_\mathrm{L}$ and the unlabeled data $\boldsymbol{X}_\mathrm{U}$ and thus can also be represented as follows:

$$\boldsymbol{X} = \{\boldsymbol{X}_\mathrm{L}, \boldsymbol{X}_\mathrm{U}\} \tag{2}$$

where $\boldsymbol{X}_\mathrm{L}$ comprises Q labeled samples such that $\boldsymbol{X}_\mathrm{L} = \{\boldsymbol{x}_{\mathrm{L}_1}, \boldsymbol{x}_{\mathrm{L}_2}, \ldots, \boldsymbol{x}_{\mathrm{L}_Q}\}$ and similarly $\boldsymbol{X}_\mathrm{U}$ comprises $P - Q$ unlabeled samples. $\boldsymbol{X}$ can be considered as a $M \times P$ matrix and represented in terms of $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_P]$. Both the set representation and the matrix representation are used in our work interchangeably without confusion. Let $\boldsymbol{Y}_\mathrm{L} = \{\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_Q\}$ denote the ground truth labels for $\boldsymbol{X}_\mathrm{L}$.

The training process aims at computing the optimal $\hat{\boldsymbol{\beta}}$ as follows:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \mathcal{O}(\boldsymbol{Y}_\mathrm{L}, C(\boldsymbol{X}_\mathrm{L}; \boldsymbol{\alpha}, \boldsymbol{\beta})) \tag{3}$$

where $\mathcal{O}(\cdot)$ is the objective function that characteries consistency between $\boldsymbol{Y}_\mathrm{L}$ and $C(\boldsymbol{X}_\mathrm{L}; \boldsymbol{\alpha}, \boldsymbol{\beta})$.

The trained classifier $C(\,\cdot\,\boldsymbol{\alpha}, \hat{\boldsymbol{\beta}})$ can be used to classify unlabeled data $\boldsymbol{X}_\mathrm{U}$. Different unlabeled samples have different confidence of being correctly classified, and the confidence is normally reflected by classification scores given by $C(\cdot\,\boldsymbol{\alpha}, \hat{\boldsymbol{\beta}})$. The high confidence classified data $\boldsymbol{X}_\mathrm{H}$ (i.e., the unlabeled data which are assigned class labels by the trained classifier with high confidence) can be selected out from the whole unlabeled data $\boldsymbol{X}_\mathrm{U}$ as follows:

$$\boldsymbol{X}_\mathrm{H} = \mathcal{R}_\gamma\left(C\left(\boldsymbol{X}_\mathrm{U}; \boldsymbol{\alpha}, \hat{\boldsymbol{\beta}}\right)\right), \quad \text{s.t. } \boldsymbol{X}_\mathrm{H} \subseteq \boldsymbol{X}_\mathrm{U} \tag{4}$$

where $\mathcal{R}_\gamma(\cdot)$ represents a ranking function which selects out the top γ percentage of data samples with high confidence. The high confidence labels assigned by the trained classifier with $\hat{\boldsymbol{\beta}}$ are computed as follows:

$$\boldsymbol{Y}_\mathrm{H} = C\left(\boldsymbol{X}_\mathrm{H}; \boldsymbol{\alpha}, \hat{\boldsymbol{\beta}}\right). \tag{5}$$

We refer to $\boldsymbol{X}_\mathrm{H}$ and $\boldsymbol{Y}_\mathrm{H}$ as high confidence classification results. More specifically, we refer to $\boldsymbol{X}_\mathrm{H}$ as guide data, and refer to the labels $\boldsymbol{Y}_\mathrm{H}$ as the guide labels for $\boldsymbol{X}_\mathrm{H}$. They are not ground truth but produced by the trained classifier with high confidence. They provide guide information for training with unlabeled data.

We refer to the classifier discussed thus far as the base classifier. A qualified base classifier is supposed to satisfy the requirement as follows.

Proposition 1: One requirement for a base classifier is that it can be randomly initialized such that its initialization is different from that of another identically structured base classifier.

Sections II-B and II-C will describe how to exploit the guide labels to train two base classifiers and establish an effective classification framework.

B. Mutual Guide Between Two Base Classifiers

We develop a mutual guide framework for training two classifiers with small data. The mutual guide framework is illustrated in Fig. 1.

The two base classifiers are initialized differently, enabling them to be able to learn different aspects from the training data.

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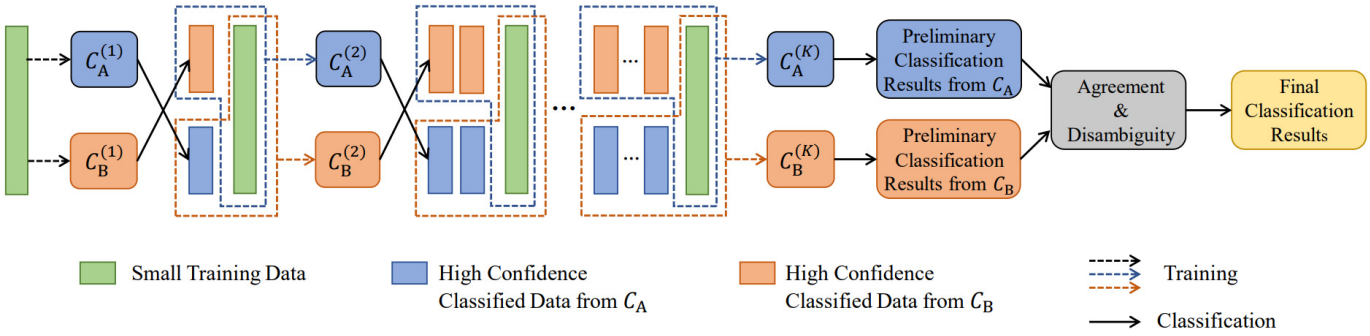


Fig. 1. Mutual guide framework for classification ($C_A^{(k)}$ and $C_B^{(k)}$ denote the base classifiers at the kth iteration).

They are trained in an iterative manner. Each iteration consists of three steps, that is: 1) training; 2) classification; and 3) augmentation. We describe the specific operations of the three steps at the kth iteration as follows.

1) Training With Incremental Labeled Data in a Mutual Guide Fashion: Let C_A and C_B denote two base classifiers with different randomly initialized parameters $\boldsymbol{\alpha}_A$ and $\boldsymbol{\alpha}_B$. Training them at the kth iteration is described as follows:

$$\hat{\boldsymbol{\beta}}_A^{(k)} = \arg\min_{\boldsymbol{\beta}_A} \mathcal{O}\left(\boldsymbol{Y}_{L+B}^{(k-1)}, C\left(\boldsymbol{X}_{L+B}^{(k-1)}; \boldsymbol{\alpha}_A, \boldsymbol{\beta}_A\right)\right) \quad (6)$$

$$\hat{\boldsymbol{\beta}}_B^{(k)} = \arg\min_{\boldsymbol{\beta}_B} \mathcal{O}\left(\boldsymbol{Y}_{L+A}^{(k-1)}, C\left(\boldsymbol{X}_{L+A}^{(k-1)}; \boldsymbol{\alpha}_B, \boldsymbol{\beta}_B\right)\right) \quad (7)$$

where $\boldsymbol{\beta}_A$ and $\boldsymbol{\beta}_B$ represent the training parameters of the base classifiers C_A and C_B, respectively. C_A and C_B can be fully expressed as $C(\cdot\ \boldsymbol{\alpha}_A, \boldsymbol{\beta}_A)$ and $C(\cdot\ \boldsymbol{\alpha}_B, \boldsymbol{\beta}_B)$ in (6) and (7), respectively. We use both the abbreviated notations and their corresponding full notations interchangeably all through our work without ambiguity. Note that $\boldsymbol{\alpha}_A$ and $\boldsymbol{\alpha}_B$ are not updated but keep the same as the their random initializations at the first iteration separately.

$\boldsymbol{X}_{L+B}^{(k-1)}$ and $\boldsymbol{Y}_{L+B}^{(k-1)}$ represent the guide data and guide labels for training the classifier C_A at the kth iteration. $\boldsymbol{X}_{L+A}^{(k-1)}$ and $\boldsymbol{Y}_{L+A}^{(k-1)}$ represent the guide data and guide labels for training the classifier C_B at the kth iteration. Especially, when $k = 1$, $\boldsymbol{X}_{L+A}^{(k-1)} = \boldsymbol{X}_{L+B}^{(k-1)} = \boldsymbol{X}_L$ and $\boldsymbol{Y}_{L+A}^{(k-1)} = \boldsymbol{Y}_{L+B}^{(k-1)} = \boldsymbol{Y}_L$.

The details for forming guide data will be presented in Sections II-B2 and II-B3. The key manipulation here is that the guide data produced by C_B at the $(k-1)$th iteration is used to train C_A at the kth iteration, and vice versa. In the same way, C_A and C_B provide guide information for training each other. One base classifier uses its representational power to encourage the other base classifier to be trained effectively.

2) Classification of Decremental Unlabeled Data in a Mutual Guide Fashion: The high confidence classified data (i.e., the new guide data) at the kth iteration are selected out as follows:

$$\boldsymbol{X}_{HA}^{(k)} = \mathcal{R}_\gamma\left(C\left(\boldsymbol{X}_{U\text{-}A}^{(k-1)}; \boldsymbol{\alpha}_A, \hat{\boldsymbol{\beta}}_A^{(k)}\right)\right) \quad (8)$$

$$\boldsymbol{X}_{HB}^{(k)} = \mathcal{R}_\gamma\left(C\left(\boldsymbol{X}_{U\text{-}B}^{(k-1)}; \boldsymbol{\alpha}_B, \hat{\boldsymbol{\beta}}_B^{(k)}\right)\right) \quad (9)$$

where $\boldsymbol{X}_{U\text{-}A}^{(k-1)}$ and $\boldsymbol{X}_{U\text{-}B}^{(k-1)}$ are the unlabeled data with respect to the classifiers C_A and C_B at the $(k-1)$th iteration. Especially, when $k = 1$, $\boldsymbol{X}_{U\text{-}A}^{(k-1)} = \boldsymbol{X}_{U\text{-}B}^{(k-1)} = \boldsymbol{X}_U$.

The guide labels of the new guide data are produced by the trained classifiers $C(\cdot\ \boldsymbol{\alpha}_A, \hat{\boldsymbol{\beta}}_A^{(k)})$ and $C(\cdot\ \boldsymbol{\alpha}_B, \hat{\boldsymbol{\beta}}_B^{(k)})$ as follows:

$$\boldsymbol{Y}_{HA}^{(k)} = C\left(\boldsymbol{X}_{HA}^{(k)}; \boldsymbol{\alpha}_A, \hat{\boldsymbol{\beta}}_A^{(k)}\right) \quad (10)$$

$$\boldsymbol{Y}_{HB}^{(k)} = C\left(\boldsymbol{X}_{HB}^{(k)}; \boldsymbol{\alpha}_B, \hat{\boldsymbol{\beta}}_B^{(k)}\right). \quad (11)$$

The new guide data and guide labels will be incorporated into the existing ones, forming augmented labeled dataset for the training at the subsequent iteration. The amounts of the unlabeled data in $\boldsymbol{X}_{U\text{-}A}^{(k-1)}$ and $\boldsymbol{X}_{U\text{-}B}^{(k-1)}$ keep decreasing with respect to the increase of the iteration k. This will be presented in the Section II-B3.

3) Augmentation of Labeled Data: The labeled data (i.e., guide data) at the kth iteration is formed by adding the new guide data obtained at kth iteration to the existing guide data at the $(k-1)$th iteration as follows:

$$\boldsymbol{X}_{L+A}^{(k)} = \boldsymbol{X}_{L+A}^{(k-1)} \cup \boldsymbol{X}_{HA}^{(k)} \quad (12)$$

$$\boldsymbol{X}_{L+B}^{(k)} = \boldsymbol{X}_{L+B}^{(k-1)} \cup \boldsymbol{X}_{HB}^{(k)}. \quad (13)$$

Accordingly, the guide labels update as follows:

$$\boldsymbol{Y}_{L+A}^{(k)} = \boldsymbol{Y}_{L+A}^{(k-1)} \cup \boldsymbol{Y}_{HA}^{(k)} \quad (14)$$

$$\boldsymbol{Y}_{L+B}^{(k)} = \boldsymbol{Y}_{L+B}^{(k-1)} \cup \boldsymbol{Y}_{HB}^{(k)}. \quad (15)$$

The unlabeled data are decremental by subtracting the new guide data

$$\boldsymbol{X}_{U\text{-}A}^{(k)} = \boldsymbol{X}_{U\text{-}A}^{(k-1)} - \boldsymbol{X}_{HA}^{(k)} \quad (16)$$

$$\boldsymbol{X}_{U\text{-}B}^{(k)} = \boldsymbol{X}_{U\text{-}B}^{(k-1)} - \boldsymbol{X}_{HB}^{(k)}. \quad (17)$$

The augmentation step increases the amounts of guide data and guide labels. It gradually enlarges the high confident training datasets with increasing iteration. This enables the continuous improvement of training effectiveness for base classifiers.

C. Final Classification Results via Agreement and Disambiguity Between the Two Base Classifiers

The mutual guide operations between the two base classifiers terminate at the Kth iteration when the amount of guide data is sufficiently large for rendering effective training.

$C(\cdot\ \boldsymbol{\alpha}_A, \hat{\boldsymbol{\beta}}_A^{(K)})$ and $C(\cdot\ \boldsymbol{\alpha}_B, \hat{\boldsymbol{\beta}}_B^{(K)})$ are the most effective classifiers over all iterations, because they are obtained by being trained with a large amount of guide data. $C(\boldsymbol{X}_U; \boldsymbol{\alpha}_A, \hat{\boldsymbol{\beta}}_A^{(K)})$

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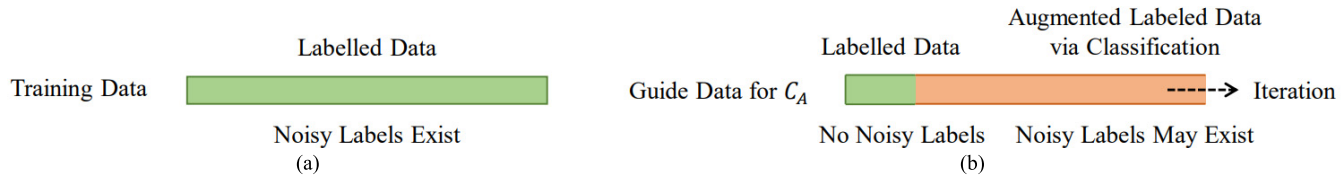


Fig. 2. Data organizations for training. (a) Co-teaching. (b) Mutual guide.

and $C(\boldsymbol{X}_{\mathrm{U}}; \boldsymbol{\alpha}_{\mathrm{B}}, \hat{\boldsymbol{\beta}}_{\mathrm{B}}^{(K)})$ give the preliminary classification results for the unlabeled data $\boldsymbol{X}_{\mathrm{U}}$ from the perspectives of C_{A} and C_{B}, respectively. The final classification result for an unlabeled sample $\boldsymbol{x} \in \boldsymbol{X}_{\mathrm{U}}$ is given with the consideration of its preliminary class labels obtained from both $C(\boldsymbol{x}; \boldsymbol{\alpha}_{\mathrm{A}}, \hat{\boldsymbol{\beta}}_{\mathrm{A}}^{(K)})$ and $C(\boldsymbol{x}; \boldsymbol{\alpha}_{\mathrm{B}}, \hat{\boldsymbol{\beta}}_{\mathrm{B}}^{(K)})$. Two situations are considered as follows.

1) Agreement: If the preliminary class labels obtained from $C(\boldsymbol{x}; \boldsymbol{\alpha}_{\mathrm{A}}, \hat{\boldsymbol{\beta}}_{\mathrm{A}}^{(K)})$ and $C(\boldsymbol{x}; \boldsymbol{\alpha}_{\mathrm{B}}, \hat{\boldsymbol{\beta}}_{\mathrm{B}}^{(K)})$ for the sample $\boldsymbol{x}$ are the same, the two classifiers have agreement on the label assignment for the sample. In this case, the final classification result for the sample is the agreed label.

2) Disambiguity: If the preliminary class labels obtained from $C(\boldsymbol{x}; \boldsymbol{\alpha}_{\mathrm{A}}, \hat{\boldsymbol{\beta}}_{\mathrm{A}}^{(K)})$ and $C(\boldsymbol{x}; \boldsymbol{\alpha}_{\mathrm{B}}, \hat{\boldsymbol{\beta}}_{\mathrm{B}}^{(K)})$ for the sample $\boldsymbol{x}$ are different, an ambiguity arises for assigning a label to the sample. We address the ambiguity by considering the label with higher confidence as the classification result for the sample.

D. Observations

Two observations about the capacity and robustness of our mutual guide framework are made as follows.

1) Training From Small Data to Augmented Labeled Data: Training solely with small data cannot render effective classifiers. To address this limitation, the mutual guide framework extends an iterative alternation between training and classification, in which the training data grow from the original small amount to augmented labeled data until a sufficient amount of data in an HSI being labeled with a certain confidence. The mutual guide framework exploits high confidence classification results as guide data to augment labeled data and thus forms larger training data. This data augmentation strategy makes the mutual guide framework resemble semi-supervised learning. However, the mutual guide framework is different from traditional semi-supervised learning. Most existing semi-supervised methods use the high confidence data produced by one classifier to retrain the classifier itself [31], [32]. In this context, the high confidence arises from too much self-confidence of the classifier and may not be reliable. In contrast, the mutual guide framework uses high confidence data produced by one classifier to train the other classifier. The high confidence is the mutual confidence from each other. The high confidence classified data are produced independently of the classifier to be trained. Therefore, the high confidence classified data in the mutual guide framework avoid the self-confidence bias arising from traditional semi-supervised learning and are much more objective and reliable for training. Though the mutual guide framework commences training with small data, the training effectiveness keeps being enhanced iteratively with the increasingly augmented labeled data. Finally, effective classifiers are obtained with the training based on the original small data and augmented labeled data. Therefore, the mutual guide framework is an effective strategy for HIC with small training data.

2) Tolerance to Intermediate Noisy Labels: The guide labels $\boldsymbol{Y}_{\mathrm{HA}}^{(k)}$ and $\boldsymbol{Y}_{\mathrm{HB}}^{(k)}$ produced at the kth iteration are considered as ground truth labels for the new guide data $\boldsymbol{X}_{\mathrm{HA}}^{(k)}$ and $\boldsymbol{X}_{\mathrm{HB}}^{(k)}$, respectively. They are involved in the training data for training C_{B} and C_{A} at the $(k+1)$th iteration. There might be wrong labels in $\boldsymbol{Y}_{\mathrm{HA}}^{(k)}$ and $\boldsymbol{Y}_{\mathrm{HB}}^{(k)}$ and we refer to them as intermediate noisy labels. The mutual guide framework addresses the intermediate noisy labels twofold. First, it manages to avoid intermediate noisy labels from being produced by the based classifiers. This is effected by two manipulations: 1) only high confidence classification results are used as guide labels and 2) the high confidence classification results are produced independently of different classifiers subject to mutual confidence. The two manipulations effectively reduce the possible existence of intermediate noisy labels in guide labels. Second, even in the situation that intermediate noisy labels arise, the two classifier training strategy exploited in the mutual guide framework neutralizes the negative effects of intermediate noisy labels to a certain degree. This observation can be validated by the empirical study of co-teaching [33], [34] that the two classifier training strategy is effective in neutralizing negative effects induced by noisy labels. However, our framework is significantly different from co-teaching. The co-teaching and mutual guide organize data for training classifiers in different ways, as illustrated in Fig. 2. The co-teaching is established in a scenario where noisy labels naturally exist in big training data. In contrast, our mutual guide framework addresses noisy labels that are intermediate by-products of the iterative training subject to small training data. Though the mutual guide and co-teaching are set up in different scenarios and address different problems, the co-teaching provides an insightful validation for the mutual guide in addressing noisy labels. Therefore, even though small amounts of noisy labels might arise at the intermediate iterations of mutual guide, they rarely affect the overall training effectiveness.

III. Implementation With SuperPCA and ELMs

This section presents a mutual guide implementation approach based on SuperPCA [18] and ELMs [13]. SuperPCA behaves as a feature extractor that establishes a feature vector $\boldsymbol{x}$ for each pixel sample in an HSI. ELMs operate as base classifiers. The overall implementation diagram is illustrated in Fig. 3, and the implementation details are presented in the following sections.

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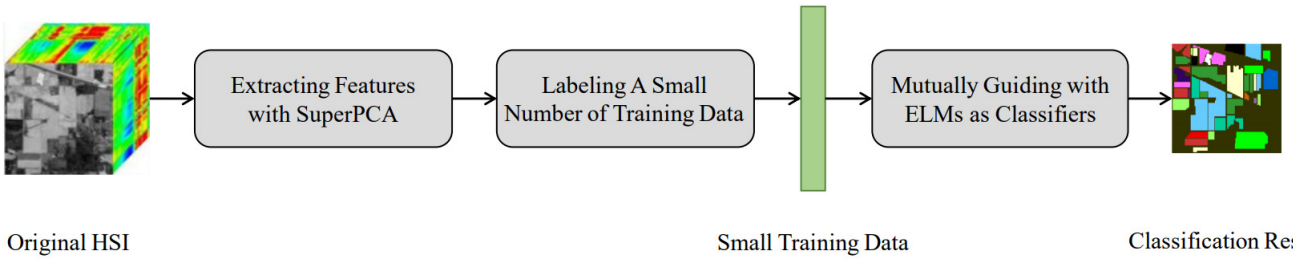


Fig. 3. Mutual guide method configured SuperPCA and ELMs for HIC.

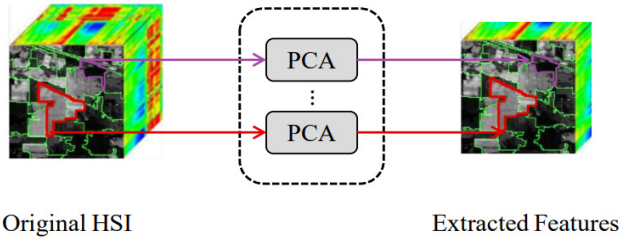


Fig. 4. HSI feature extraction based on SuperPCA.

A. SuperPCA as the Feature Extractor

Let $\mathcal{S} \in \mathbb{R}^{D\times E\times G}$ represent an HSI cube. $P = D \times E$ represents the spatial size and G represents the spectral depth. We reform $\mathcal{S} \in \mathbb{R}^{D\times E\times G}$ into a matrix $\boldsymbol{S} \in \mathbb{R}^{P\times G}$, which is a more convenient form for further processing.

SuperPCA is explored to superpixelwise feature extraction, as shown in Fig. 4. First, an HSI is divided into I homogeneous regions through superpixel segmentation based on [35], and the homogeneous regions are represented by matrices $\boldsymbol{S} = [\boldsymbol{S}_1, \boldsymbol{S}_2, \ldots, \boldsymbol{S}_I]$. Second, PCA is employed to extract low-dimension features in each homogeneous region. Let $\boldsymbol{T}_i$ denote the transformation matrix of the ith $(1 \le i \le I)$ region $\boldsymbol{S}_i$. The PCA establishes the $G \times M$ linear transformation matrix $\hat{\boldsymbol{T}}_i$ by optimizing the following objective function:

$$\hat{\boldsymbol{T}}_i = \arg\max_{\boldsymbol{T}_i} \mathrm{Tr}\left(\boldsymbol{T}_i^\top \mathrm{Cov}(\boldsymbol{S}_i)\boldsymbol{T}_i\right) \tag{18}$$

where $\mathrm{Cov}(\cdot)$ represents the covariance matrix, and $\mathrm{Tr}(\cdot)$ represents the trace. The original G-dimensional data $\boldsymbol{S}_i$ is transformed to M-dimensional data $\boldsymbol{X}_i$ as follows:

$$\boldsymbol{X}_i = \hat{\boldsymbol{T}}_i \boldsymbol{S}_i. \tag{19}$$

The extracted features for all regions form a low-dimensional HSI representation $\boldsymbol{X} = [\boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_I]$ and can also be expressed as a set of samples $\boldsymbol{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_P\}$.

B. ELMs as the Base Classifiers

An ELM is a single hidden layer neural network with strong learning ability, which only has an input layer, a hidden layer, and an output layer. The input weight and the hidden bias of the ELM are randomly generated. This property enables an ELM to satisfy the requirement given in Proposition 1. Therefore, it is a qualified base classifier. According to the formulations in Section II, we equip a base classifier with an ELM as follows:

$$C(\boldsymbol{X}_{\mathrm{L}}; \boldsymbol{\alpha}, \boldsymbol{\beta}) = \left[\boldsymbol{\phi}\left(\boldsymbol{X}_{\mathrm{L}}^\top \boldsymbol{W} + \mathbf{1}\boldsymbol{v}^\top\right)\right]\boldsymbol{\beta}, \text{ where } \boldsymbol{\alpha} = \{\boldsymbol{W}, \boldsymbol{v}\} \tag{20}$$

where $\boldsymbol{W} = [\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_J]$, $\boldsymbol{\beta} = [\boldsymbol{\beta}_1, \boldsymbol{\beta}_2, \ldots, \boldsymbol{\beta}_J]$ and $\boldsymbol{v} = [v_1, v_2, \ldots, v_J]^\top$ denote the input weight, the output weight, and the hidden bias of the ELM, respectively. Additionally, $\boldsymbol{W}$ and $\boldsymbol{v}$ are random parameters and form $\boldsymbol{\alpha} = \{\boldsymbol{W}, \boldsymbol{v}\}$ which is responsible for the random initialization for the base classifier. $\boldsymbol{\phi}(\cdot)$ is the activation function, which is required to be infinitely differentiable. $\mathbf{1} = [1, 1, \ldots, 1]^\top$ is a unit column vector contains Q ones.

The objective function for training the ELM is given as follows:

$$\begin{aligned}\hat{\boldsymbol{\beta}} &= \arg\min_{\boldsymbol{\beta}} \mathcal{O}(\boldsymbol{Y}_{\mathrm{L}}, C(\boldsymbol{X}_{\mathrm{L}}; \boldsymbol{\alpha}, \boldsymbol{\beta})) \\ &= \arg\min_{\boldsymbol{\beta}} \left\| \left[\boldsymbol{\phi}\left(\boldsymbol{X}_{\mathrm{L}}^\top \boldsymbol{W} + \mathbf{1}\boldsymbol{v}^\top\right)\right]\boldsymbol{\beta} - \boldsymbol{Y}_{\mathrm{L}} \right\|. \end{aligned} \tag{21}$$

Specifically, $\hat{\boldsymbol{\beta}}$ is solved according to the smallest norm least-squares solution of a linear system as follows:

$$\hat{\boldsymbol{\beta}} = \left[\boldsymbol{\phi}\left(\boldsymbol{X}_{\mathrm{L}}^\top \boldsymbol{W} + \mathbf{1}\boldsymbol{v}^\top\right)\right]^\dagger \boldsymbol{Y}_{\mathrm{L}} \tag{22}$$

where † represents the Moore-Penrose generalized inverse.

The relations in (21) and (22) are specific implementations of that in (3). According to this configuration, the mutual guide training presented in Section II-B and classification presented in Section II-C can be easily implemented.

IV. Experimental Results

We empirically evaluate our mutual guide framework by making qualitative and quantitative comparisons with several state-of-the-art methods on four public HSI datasets, i.e., Indian Pines (IP), Kennedy Space Center (KSC), University of Pavia (UP), and Salinas (SA).

A. Experimental Datasets

Four public HSI datasets captured by different sensors and covering different areas are employed to validate the effectiveness of our mutual guide framework. The descriptions of them are briefly given below.

The first dataset is IP, which was captured by the sensor named Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) over the IP test site in northwestern Indiana, USA. It possesses 145×145 spatial pixels and 200 available

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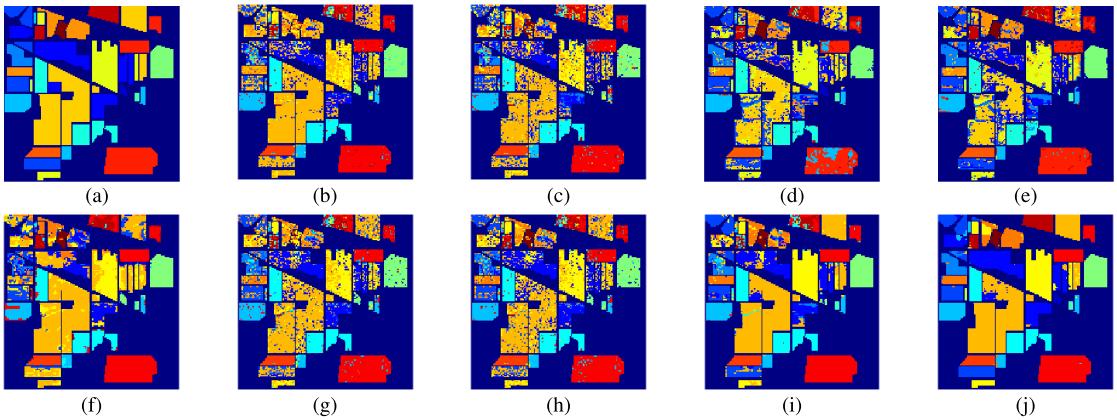


Fig. 5. Visual illustration of classification on the IP dataset. (a) Ground truth. (b) SVM. (c) ELM. (d) 2DCNN. (e) 3DCNN [23]. (f) CDCNN [25]. (g) GCN [29]. (h) miniGCN [30]. (i) FDSI [36]. (j) Mutual guide.

reflectance spectral bands ranging from 0.4 to 2.5 μm with ground sample distance (GSD) of 20 m. There are 10 249 labeled pixels with 16 different land coverings (classes) in the ground truth of the IP dataset.

The second dataset is KSC, which was captured by the AVIRIS sensor over the KSC, Florida, USA. It possesses 512 $\times$ 614 spatial pixels and 176 available reflectance spectral bands ranging from 0.4 to 2.5 μm with GSD of 18 m. There are 5211 labeled pixels with 13 classes in the ground truth of the KSC dataset.

The third dataset is UP, which was captured by the sensor named Reflective Optics Spectrographic Imaging System (ROSIS) over the urban area of the UP, Italy. It possesses 610 $\times$ 340 spatial pixels and 103 available reflectance spectral bands ranging from 0.43 to 0.86 μm with GSD of 1.3 m. There are 42 776 labeled pixels with nine classes in the ground truth of the UP dataset.

The fourth dataset is SA, which was captured by the AVIRIS sensor over the SA Valley, California, USA. It possesses 512 $\times$ 217 spatial pixels and 184 available reflectance spectral bands ranging from 0.4 to 2.5μm with GSD of 3.7 m. There are 54 129 labeled pixels with 16 classes in the ground truth of the SA dataset.

B. Experimental Configurations

The classification results of our mutual guide framework are assessed via quantitative comparisons with eight state-of-the-art methods including SVM, ELM, 2DCNN, 3DCNN [23], CDCNN [25], GCN [29], miniGCN [30], and FDSI [36]. In order to objectively and persuasively illustrate the effectiveness of our mutual guide method, we make detailed experimental configurations, which are delineated below.

There are 5% samples randomly selected from the total data for training. The class labels for the training samples are supposed to be known. The remaining 95% samples are for testing. In each iteration, the classifier C_{A} selects high confidence classification results (which are 0.5% of the total data in amount) as new labeled samples to augment the labeled dataset. The new labels are not given as prior knowledge but predicted by the classifier C_{A}. The classifier C_{B} is trained based on the augmented labeled dataset provided by the classifier C_{A} in each iteration. In the same manner, the classifier C_{B} maintains an augmented labeled dataset for retraining the classifier C_{A} in each iteration. Ten iterations are conducted.

The proposed mutual guide and the comparison methods use different types of hyperparameter settings, because they are architecturally different. The proposed mutual guide is not necessarily a deep learning method but implemented in terms of SuperPCA and ELMs in our work. The hyperparameter setting for our method is the proportion of high confidence data and the number of iterations. The proportion of high confidence data is set to 0.5%, which is empirically the most favorable hyperparameter setting (see Section IV-F for details). The number of iterations is set to 10, which is sufficient to achieve good performance. The hyperparameter settings for the comparison methods are as follows. The number of epochs is set to 200, the number of iterations is set to 10, and the value of the initial learning rate is set to 0.001. The batch size and patch size are set to 16 $\times$ 16 and 3 $\times$ 3, respectively.

We adopt three critical metrics of hyperspectral classification to measure the classification performance, i.e., overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa). Specifically, OA measures the percentage of all correctly classified pixels over the total pixels of all land cover classes. AA is the average of recalls over all classes. One recall is the percentage of the correctly classified pixels in a land cover class over all pixels in the land cover class. The Kappa coefficient is an indicator used for consistency testing and can also be used to measure the effectiveness of classifications. It represents the percentage of errors that are reduced by classifications compared to completely random classifications.

C. Qualitative Evaluations

Figs. 5–8 visually demonstrate the classification maps acquired by various methods using the four public HSI datasets. We use false-color images as classification maps and exhibit them for illustrating the ground truth and different classification results.

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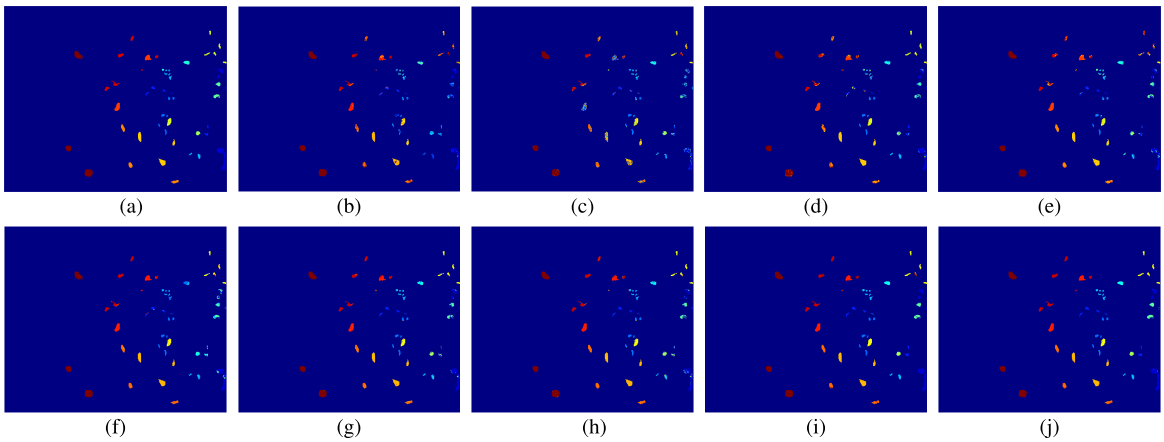


Fig. 6. Visual illustration of classification on the KSC dataset. (a) Ground truth. (b) SVM. (c) ELM. (d) 2DCNN. (e) 3DCNN [23]. (f) CDCNN [25]. (g) GCN [29]. (h) miniGCN [30]. (i) FDSI [36]. (j) Mutual guide.

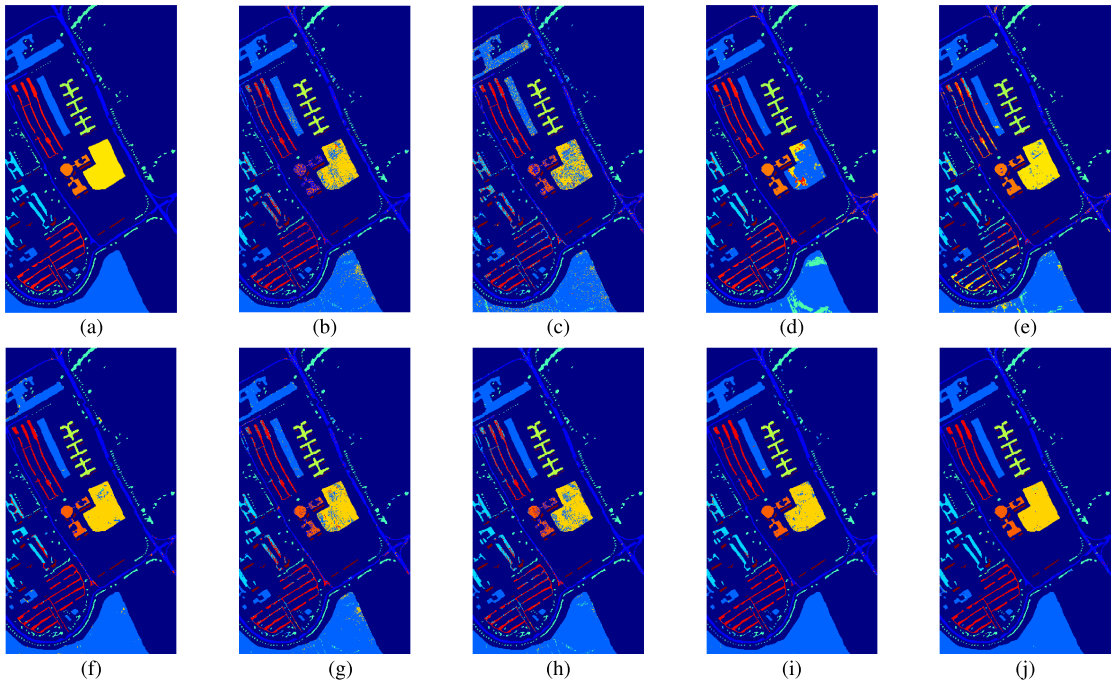


Fig. 7. Visual illustration of classification on the UP dataset. (a) Ground truth. (b) SVM. (c) ELM. (d) 2DCNN. (e) 3DCNN [23]. (f) CDCNN [25]. (g) GCN [29]. (h) miniGCN [30]. (i) FDSI [36]. (j) Mutual guide.

With limited learning capacities, shallow learning methods such as SVM and ELM suffer from serious scatter pepper-noise alike errors on the classification maps. Object edge discrimination by traditional deep learning methods such as 2DCNN and 3DCNN is confusing and inaccurate because the spatial context information is not comprehensively exploited for these edge pixels. CDCNN, which is a CNN-based deep learning method with contextual information, is superior to the two traditional deep learning methods in terms of object edge discrimination. The classification maps of GCN are unsatisfactory. In particular, in the classification map of the IP dataset, GCN induces serious scatter pepper-noise alike errors. MiniGCN, which allows the training of large-scale GCNs in a mini-batch fashion, surpasses GCN while saving computing resources, but there are still some misclassifications. FDSI, which is based on dual spatial information fusion, produces smoother classification maps and achieves better classification performance, but there is still room for improvement. Compared with these methods, our mutual guide classification maps not only have clearer edges but also exhibit less misclassification.

D. Quantitative Evaluations

We conduct ten runs of the mutual guide method and obtain the average quantitative results. Specifically, the training data for each run are randomly selected. Tables I–IV report the quantitative comparison results, which are consistent with the classification maps illustrated in Figs. 5–8. Each table row associated with a class gives recalls regarding the class

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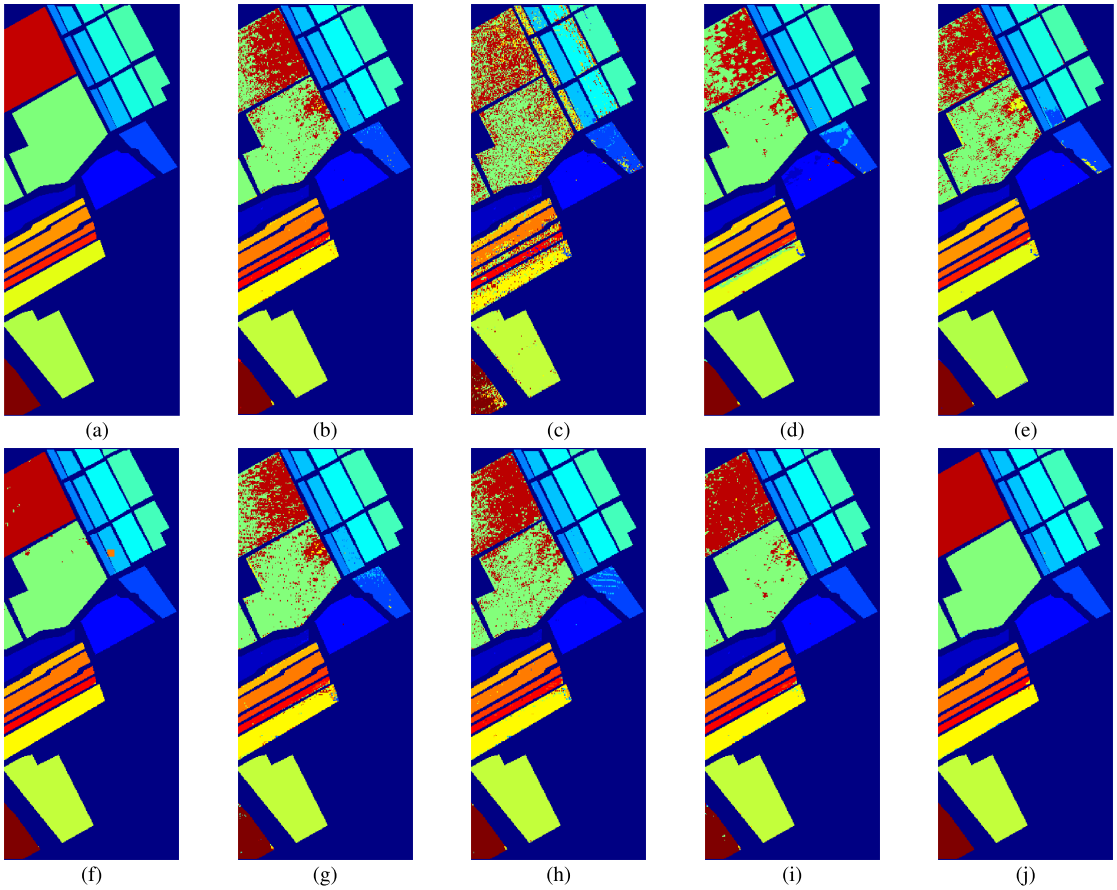


Fig. 8. Visual illustration of classification on the SA dataset. (a) Ground truth. (b) SVM. (c) ELM. (d) 2DCNN. (e) 3DCNN [23]. (f) CDCNN [25]. (g) GCN [29]. (h) miniGCN [30]. (i) FDSI [36]. (j) Mutual guide.

obtained from different methods. The bottom four table rows report OA, AA, and Kappa obtained from the different methods, separately. Experimental evaluations are summarized as follows.

1) Our Mutual Guide Method Outperforms All Comparison Methods: The performance of our mutual guide framework on the four datasets is not only significantly better than the shallow learning methods but also superior to the deep learning methods. Even compared with the state-of-the-art CDCNN that integrates contextual information and the FDSI that fuses dual spatial information, our mutual guide yields more remarkable classification performance. Specifically, the OA of our mutual guide method on the IP, KSC, UP, and SA datasets is 96.83%, 97.67%, 98.18%, and 99.67%, indicating the superiority of our mutual guide framework in HIC with small training data. The empirical effectiveness of our mutual guide over the comparison methods is described in detail in the following paragraphs.

2) Our Mutual Guide Method Outperforms Shallow Learning Methods: The shallow learning methods avoid the complicated training of deep models to a certain extent. However, their straightforward network structure and the shortage of spatial information lead to unsatisfactory classification performance. The OA of our mutual guide method on the four datasets outperforms SVM by 29.93%, 29.45%, 9.61%, and 7.99%, and surpasses ELM by 33.45%, 34.62%, 17.25%, and 24.51%. The state-of-the-art SuperPCA is used to perform feature extraction in implementing our mutual guide framework. By dividing an HSI into homogeneous regions, the spatial context information is integrated into the unsupervised dimensionality reduction of superpixel segmentation. It alleviates the insufficient use of spatial information with shallow learning methods. The essential drawback of these shallow learning methods is that their learning capacities are generally limited. In contrast, our mutual guide method composites two shallow ELMs to a strong classification model with an iterative training strategy, which gradually enhances the model representational power. Therefore, even though our ELM-based mutual guide exploits shallow structures as base classifiers, it yields significantly better classification performance than the traditional shallow learning methods.

3) Our Mutual Guide Method Outperforms Deep Learning Methods: Deep learning methods are data-driven machine learning methods that depend on the quantity and quality of labeled data to a certain extent. With only 5% of labeled data selected randomly for training, the deep learning methods are prone to overfitting, which seriously affects the classification accuracy. The OA of our mutual guide method exceeds 3DCNN by 26.52%, 15.51%, 7.03%, and 9.23% on the four datasets. Even though 3DCNN achieves a better performance than 2DCNN, it has obvious shortcomings in processing the HIC with small training data. Compared with deep learning

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TABLE I
CLASSIFICATION ACCURACY ON THE IP DATASET (%)

Class	SVM	ELM	2DCNN	3DCNN [23]	CDCNN [25]	GCN [29]	miniGCN [30]	FDSI [36]	Mutual Guide
1	0	45.45	**100**	64.52	39.53	13.64	20.45	50.00	**100**
2	54.39	49.34	36.68	52.57	69.18	62.37	64.42	70.28	**94.14**
3	29.04	39.65	71.80	69.32	55.10	50.00	51.77	81.94	**93.06**
4	23.42	38.74	79.51	92.49	61.20	53.15	44.59	66.67	**97.75**
5	86.40	78.95	89.58	86.63	91.92	79.61	85.96	92.76	**98.03**
6	96.06	92.71	94.84	92.94	91.32	96.21	94.90	99.56	**100**
7	64.00	52.00	**100**	**100**	48.83	56.00	68.00	92.00	96.00
8	92.86	88.10	95.96	96.86	89.20	90.69	90.48	**100**	99.57
9	11.11	11.11	**100**	**100**	64.13	11.11	11.11	38.89	**100**
10	41.84	61.23	80.00	69.16	57.24	66.37	72.29	92.66	**98.69**
11	96.68	70.03	54.85	52.36	73.27	72.50	68.23	97.65	**98.04**
12	22.34	40.43	69.52	75.43	48.61	39.36	44.50	65.07	**83.16**
13	95.90	95.90	98.27	**100**	92.59	96.92	97.95	99.49	99.49
14	93.72	91.88	72.59	87.93	89.25	92.55	90.28	99.33	**99.66**
15	39.51	25.07	66.67	79.50	79.14	44.41	44.96	43.32	**98.61**
16	0	56.04	**100**	96.55	89.48	26.37	85.71	94.51	97.80
OA	66.90	63.38	68.05	70.31	70.58	70.24	70.83	87.32	**96.83±0.15**
AA	52.33	58.54	82.08	82.27	71.25	59.52	64.66	80.26	**97.06±0.25**
Kappa	61.15	60.24	64.10	66.54	66.54	65.87	66.73	85.43	**96.38±0.17**

TABLE II
CLASSIFICATION ACCURACY ON THE KSC DATASET (%)

Class	SVM	ELM	2DCNN	3DCNN [23]	CDCNN [25]	GCN [29]	miniGCN [30]	FDSI [36]	Mutual Guide
1	94.69	69.62	92.12	86.31	96.35	95.27	91.33	96.45	**97.63**
2	58.16	70.72	74.03	90.04	81.18	86.42	81.89	95.36	**96.98**
3	58.75	76.32	90.57	83.61	69.01	91.80	88.28	97.12	**98.33**
4	23.93	46.88	56.67	27.08	41.97	64.68	48.41	**99.57**	95.30
5	0	43.66	47.71	58.17	53.16	54.66	48.45	**91.35**	86.18
6	5.14	16.59	47.71	56.88	66.79	72.49	73.36	95.33	**99.07**
7	0.99	80.21	61.00	71.00	59.18	87.62	83.81	**100**	**100**
8	30.77	59.75	89.02	62.20	81.39	90.49	91.18	95.87	**99.52**
9	87.70	65.65	**98.99**	94.13	88.23	96.73	97.88	95.00	98.19
10	65.17	59.05	69.53	87.76	93.55	91.09	95.05	94.41	**95.51**
11	87.87	0.78	93.37	95.49	97.68	93.79	95.23	98.20	**99.50**
12	63.62	67.54	61.51	80.13	91.56	89.07	85.88	85.86	**99.79**
13	99.89	97.02	93.87	98.18	98.03	99.78	99.03	**100**	99.10
OA	68.22	63.05	81.70	82.16	84.45	90.31	88.49	96.36	**97.67±0.27**
AA	52.05	57.98	75.11	76.23	78.31	85.68	83.06	95.81	**97.21±0.34**
Kappa	64.20	58.90	79.58	80.16	82.68	89.21	87.19	95.94	**97.40±0.30**

methods, our mutual guide method trains two ELMs that do not require numerous labeled training data in an iterative manner, thereby yielding better classification performance. It can be observed that the classification performance is not directly determined by the depth of a model in the situation of small training data.

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TABLE III
CLASSIFICATION ACCURACY ON THE UP DATASET (%)

Class	SVM	ELM	2DCNN	3DCNN [23]	CDCNN [25]	GCN [29]	miniGCN [30]	FDSI [36]	Mutual Guide
1	92.68	80.93	89.38	92.16	93.94	91.09	92.35	**97.65**	97.45
2	96.10	88.75	92.20	95.67	98.44	97.59	98.70	99.81	**99.90**
3	70.86	56.29	92.73	85.99	81.89	77.18	87.32	92.41	**99.08**
4	88.74	77.08	**98.85**	98.04	97.77	88.33	96.51	80.67	90.18
5	99.84	99.05	**100**	99.91	99.79	99.29	99.92	62.29	97.94
6	77.61	60.23	16.71	94.34	91.83	82.70	79.85	**99.84**	98.72
7	34.23	55.28	97.61	95.30	95.57	75.79	68.05	96.77	**99.21**
8	80.50	76.81	93.80	51.95	90.67	86.09	78.81	96.62	**98.94**
9	99.22	96.23	**100**	99.73	97.46	98.56	**100**	99.67	94.57
OA	88.57	80.93	84.00	91.15	95.01	91.58	92.19	96.48	**98.18±0.09**
AA	82.20	77.27	86.81	90.35	94.15	88.51	89.06	95.08	**97.54±0.16**
Kappa	84.68	74.65	77.98	88.17	93.40	88.76	89.56	95.29	**97.59±0.12**

TABLE IV
CLASSIFICATION ACCURACY ON THE SA DATASET (%)

Class	SVM	ELM	2DCNN	3DCNN [23]	CDCNN [25]	GCN [29]	miniGCN [30]	FDSI [36]	Mutual Guide
1	99.58	98.23	95.20	99.89	98.49	98.54	99.32	99.53	**100**
2	99.89	99.18	99.32	99.74	94.98	99.92	99.75	99.97	**100**
3	97.55	90.25	99.68	95.10	96.23	91.32	87.99	99.73	**100**
4	98.78	2.05	99.62	**99.75**	99.03	98.71	97.87	99.01	99.62
5	98.90	92.03	77.80	87.69	98.92	98.08	99.45	99.18	**99.53**
6	99.66	78.56	**99.95**	98.99	99.86	99.66	99.89	99.73	99.89
7	**99.76**	98.32	98.38	99.41	99.04	99.62	**99.76**	99.53	99.44
8	85.59	66.96	94.28	79.69	84.76	85.33	80.98	95.04	**99.83**
9	99.91	96.84	99.64	98.68	99.64	**99.97**	99.90	99.73	**99.97**
10	95.16	60.87	83.08	96.36	95.22	91.34	92.46	95.77	**97.85**
11	94.42	80.06	88.67	98.27	95.76	88.33	97.61	**99.60**	**99.60**
12	99.78	80.31	98.91	**100**	99.20	99.56	98.90	**100**	**100**
13	97.47	4.14	**100**	99.16	98.24	98.97	98.85	99.54	98.05
14	92.86	81.31	89.48	97.59	98.04	89.04	92.66	94.62	**98.24**
15	66.96	54.97	59.25	72.92	81.03	63.94	76.63	91.70	**99.97**
16	98.44	68.90	98.14	99.32	99.66	97.86	99.65	98.32	**100**
OA	91.68	75.16	90.21	90.44	92.58	90.49	91.56	97.24	**99.67±0.03**
AA	95.29	72.06	92.59	95.16	96.13	93.76	95.04	98.19	**99.47±0.06**
Kappa	90.73	72.14	89.28	89.31	91.72	89.39	90.60	96.93	**99.63±0.03**

4) Our Mutual Guide Method Outperforms the Contextual Methods: CDCNN is a variant of CNN-based HIC. It integrates the spectral-spatial relationships of adjacent individual pixels to interact with local context information. With the interaction of contextual information, CDCNN outperforms the traditional deep learning methods, i.e., 2DCNN and 3DCNN, especially when classifying the KSC, UP, and SA datasets. Additionally, compared with our mutual guide method, the superiority of CDCNN has not been clearly demonstrated on the four datasets. The OA of it is 26.25%, 13.22%, 3.17%, and 7.09% lower than our mutual guide method, respectively. SuperPCA encodes spatial information for the mutual guide in the preprocessing stage, making the mutual guide not employ additional networks to extract

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contextual information. CDCNN needs to deepen the network when integrating contextual information, and this would intensify the overfitting of the deep network subject to small training data.

5) Our Mutual Guide Method Outperforms the Semi-Supervised Learning Methods: Semi-supervised learning methods can effectively take advantage of the information in unlabeled data to alleviate the insufficiency of labeled data, thereby improving classification accuracy. As we observe, the classification results of the semi-supervised method GCN on the four datasets are significantly better than 2DCNN and on a par with 3DCNN. MiniGCN, which trains large-scale GCNs in a mini-batch fashion, also achieves a remarkable classification performance. The MiniGCN classification results are significantly superior to the two traditional CNN-based methods and the traditional GCN on the IP, UP, and SA datasets. But its superiority has not been clearly demonstrated on the KSC dataset. Compared with our mutual guide method, the performance of miniGCN is unsatisfactory. The OA of it is 26.00%, 9.18%, 5.99%, and 8.11% lower than our mutual guide method on the four datasets, separately. Semi-supervised learning methods focus on using pseudo-labels to retrain classification models. However, the capacity of these methods to counteract the undesirable influence of noisy labels is limited. Our mutual guide method has tolerance to intermediate noisy labels, so that its classification performance does not degrade with respect to the increase of the training iteration.

6) Our Mutual Guide Method Outperforms the Fusion Methods: FDSI is a HIC framework using the fusion of dual spatial information. Specifically, dual spatial information is constructed by preprocessing feature extraction and post-processing space optimization. The classification performance of FSDI has obvious superiority compared to other methods but is still inferior to our mutual guide method, underperforming by 9.51%, 1.31%, 1.70%, and 2.43% on the four datasets, separately. There are dual deficiencies regarding feature extraction and model structure that lead to the unsatisfactory performance of the FDSI. In terms of feature extraction, FSDI uses traditional PCA, while our mutual guide method uses SuperPCA. Different from the traditional PCA that is based on the entire image, SuperPCA considers the diversity of different homogeneous regions and integrates spatial contextual information. In terms of the model structure, FSDI obtains the final result by weighting the classification results based on preprocessing feature extraction and the classification results based on post-processing space optimization. Determining suitable weights raises the difficulty for accurate classification. In contrast, the mutual guide operates in an iterative manner without the troublesome weighting scheme.

E. Iterative Performance Analysis

We investigate the performance trend of the mutual guide method with respect to iteration. At the same time, we also investigate the benefit of the mutual guide method over the strategy of using pseudo-labels generated by one classifier to retrain the classifier itself. The performance in terms of OA, AA, and Kappa at ten consecutive iterations is presented in Fig. 9. The curves with respect to A and B show the performance of the mutual guide classifiers C_{A} and C_{B}, respectively. The curve with respect to A′ (or B′) shows the performance of the classifier C_{A} (or C_{B}) which is trained with the pseudo-labels generated by itself. Different random initializations can make the performance of the two classifiers (with the same structure and using the same original training data) different at the first iteration. With the increase of training iteration, the two classifiers (no matter mutual guide or trained by self-provided pseudo-labels) gradually improve the classification accuracy.

Our method is implemented on a computing platform with an Intel i5-4570 CPU core at 3.2 Hz by using MATLAB. The training time on the IP, KSC, UP, and SA datasets is 31, 37, 53, and 213 s, respectively. It is clear that even if iterative training is performed, very little time is consumed. This is because our mutual guide method is implemented in terms of ELMs that do not have deep architecture. Therefore, it does not cost much computing time. The following four observations are made.

1) Trends of the Three Critical Metrics Maintain Consistency: The three critical metrics represent the classification performance, and empirically results show that they change simultaneously. Specifically, with respect to the IP dataset, the value of OA of the mutual guide classifier C_{B} (the curve B) from the second to third iteration decreases by an appropriate amount, and the corresponding values of AA and Kappa decrease. Interestingly, when the value of OA changes to increase from the third to fourth iteration, and the corresponding values of AA and Kappa also change to increase.

2) Responses of the Two Classifiers on Sample Quality Are Consistent: Although the learning abilities of the two classifiers are different, as high confidence classified data are continuously added in iterations, their responses are consistent. Specifically, from the first to fourth iteration of the KSC dataset, the classification accuracy of the two mutual guide classifiers C_{A} and C_{B} (the curves A and B) continues to increase, indicating that the two classifiers continue to increase robustness under the augmentation of high confidence samples. From the fourth to sixth iteration, both classifiers undergo declining changes, suggesting that there may be considerable incorrect samples in the selection of high confidence classified data, resulting in relatively ineffective classification performance. In subsequent iterations, their classification accuracy has a relatively stable improvement, suggesting that the robustness of the two classifiers has been continuously enhanced.

3) Our Mutual Guide Method Is Effective in the HIC With Small Training Data: Although the accuracy of the two classifiers occasionally decreases in the iterative process, their overall classification performance keeps increasing for many iterations, suggesting the effectiveness and robustness of the iterative training. Even after ten iterations, the OA, AA, and Kappa of the mutual guide classifier C_{A} (the curves A) still increase by 1.42%, 2.03%, and 1.89%, respectively, and the OA, AA, and Kappa of the mutual guide classifier C_{B} (the curves B) increase by 1.30%, 1.70%, and 1.73%, respectively, on the UP dataset. This validates that the mutual guide is capable of training with small data and maintaining growing representational power in an iterative manner.

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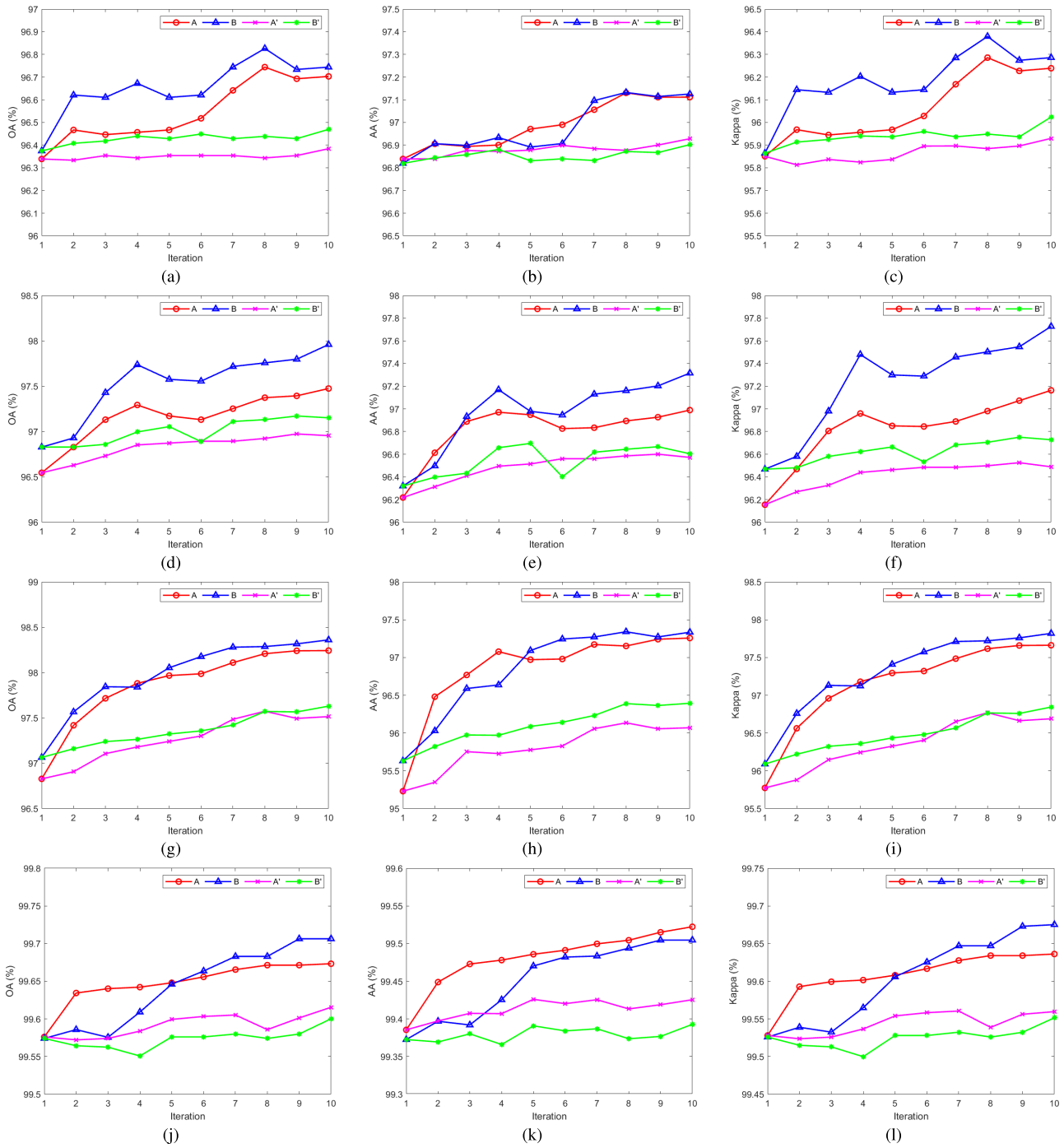


Fig. 9. Performance of our mutual guide method with respect to iterations. (a) OA on the IP dataset. (b) AA on the IP dataset. (c) Kappa on the IP dataset. (d) OA on the KSC dataset. (e) AA on the KSC dataset. (f) Kappa on the KSC dataset. (g) OA on the UP dataset. (h) AA on the UP dataset. (i) Kappa on the UP dataset. (j) OA on the SA dataset. (k) AA on the SA dataset. (l) Kappa on the SA dataset.

4) Self-Confidence Bias Is Neutralized By Mutual Guide: The curve with respect to A′ (or B′) shows the performance of the classifier C_{A} (or C_{B}) which is iteratively trained with the incrementally increasing pseudo-labels provided by itself. The pseudo-labels are determined in terms of high confidence classification results. In this scenario, the high confidence classification results arise from too much of the classifier's self-confidence and may not be reliable. The classification results which are considered by the classifier itself as correct labels may be incomprehensively wrong and would in turn induce training bias to itself, which limits the training effectiveness. In contrast, in the mutual guide framework, the high confidence classification results for training one classifier are produced independently of the other classifier. The independence of training apart from its own classification results avoids the self-confidence bias to a certain degree. Take the curves A and A′ for example. It should be noted that the curves A and A′ show the classification performance of

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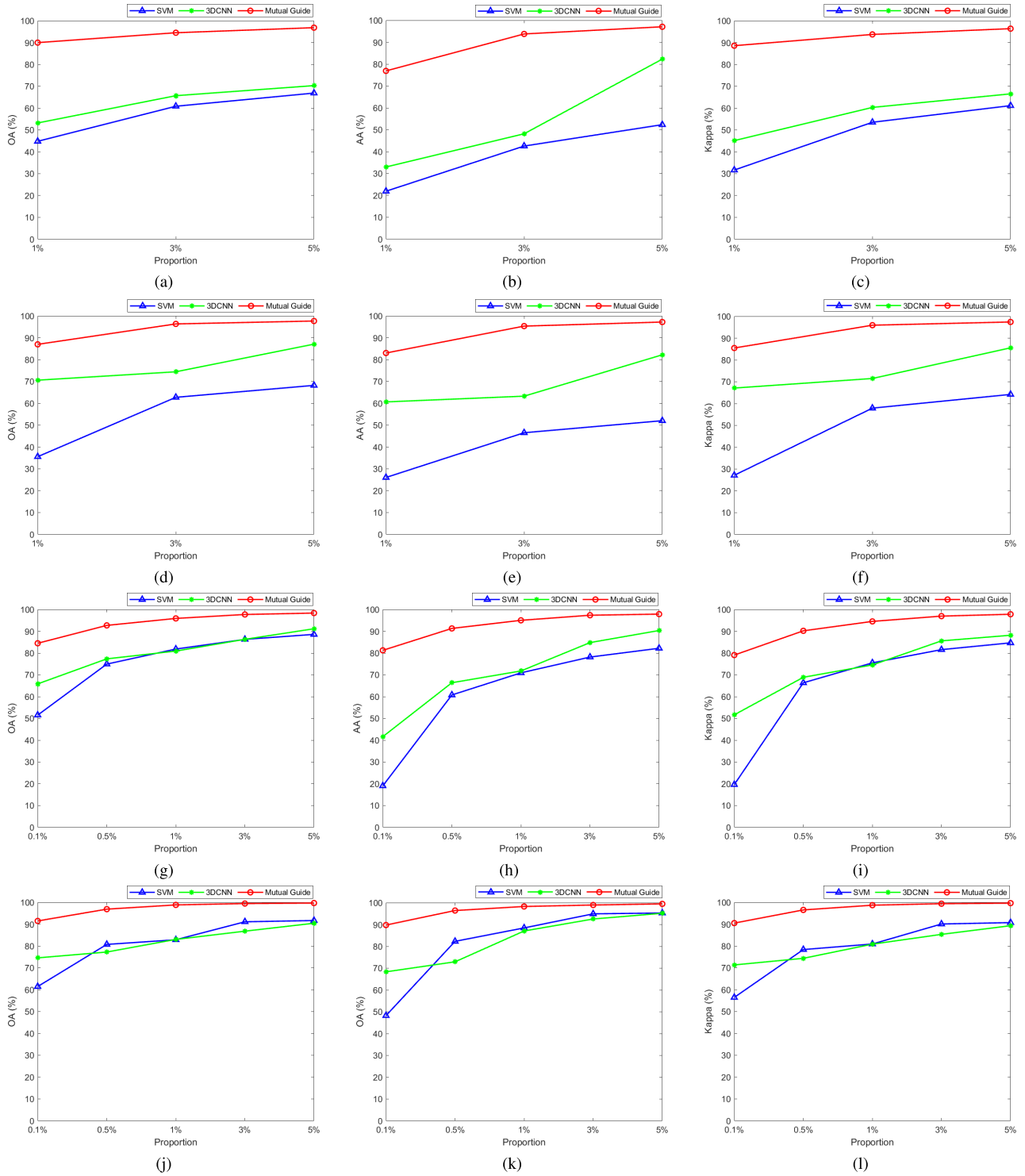


Fig. 10. Performance with different proportions of the training data. (a) OA on the IP dataset. (b) AA on the IP dataset. (c) Kappa on the IP dataset. (d) OA on the KSC dataset. (e) AA on the KSC dataset. (f) Kappa on the KSC dataset. (g) OA on the UP dataset. (h) AA on the UP dataset. (i) Kappa on the UP dataset. (j) OA on the SA dataset. (k) AA on the SA dataset. (l) Kappa on the SA dataset.

C_{A} with the same model architecture and the same model initialization. The only difference is the training strategy. The curve A for C_{A} is the obtained subject to mutual guide with the classifier C_{B}. The curve A′ for C_{A} is obtained subject to training with the pseudo-labels provided by C_{A} itself. From every subfigure in Fig. 9, we observe that the mutual guide classifiers show better performance, validating that the mutual guide framework has the capability of neutralizing self-confidence bias.

F. Ablation Studies

We test the performance of our mutual guide method by conducting experiments that progressively change the proportion of the training data and qualitatively making comparisons

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TABLE V
PERFORMANCE WITH DIFFERENT THRESHOLDS OF THE HIGH CONFIDENCE DATA (%)

Dataset	Metric	0.1%	0.3%	0.5%	0.7%	0.9%
IP	OA	96.72±0.23	96.77±0.21	**96.83±0.15**	96.63±0.13	96.58±0.21
	AA	96.85±0.25	96.93±0.26	**97.06±0.25**	96.79±0.25	96.58±0.25
	Kappa	96.25±0.26	96.32±0.23	**96.38±0.17**	96.15±0.15	96.10±0.25
KSC	OA	96.34±0.75	97.18±0.13	**97.67±0.27**	97.34±0.42	97.32±0.43
	AA	95.78±1.08	96.59±0.18	**97.21±0.34**	96.86±0.68	96.72±0.64
	Kappa	95.92±0.83	96.86±0.14	**97.40±0.30**	97.04±0.47	97.02±0.48
UP	OA	97.73±0.25	98.06±0.13	98.18±0.09	**98.31±0.04**	98.27±0.08
	AA	97.00±0.26	97.41±0.20	97.54±0.16	**97.65±0.15**	97.64±0.14
	Kappa	96.98±0.21	97.41±0.18	97.59±0.12	**97.75±0.07**	97.70±0.10
SA	OA	99.63±0.04	99.65±0.02	**99.67±0.03**	99.66±0.02	99.65±0.01
	AA	99.45±0.06	99.46±0.06	**99.47±0.06**	99.46±0.06	99.45±0.06
	Kappa	99.58±0.04	99.61±0.02	**99.63±0.03**	99.62±0.03	99.61±0.01

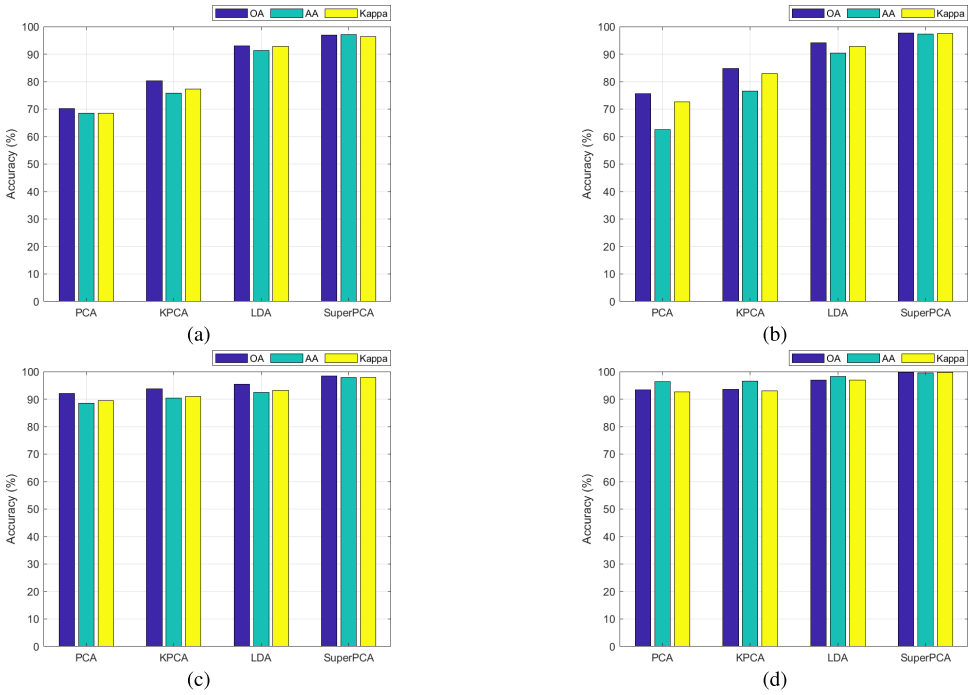


Fig. 11. Performance with different feature extraction methods. (a) Performance on the IP dataset. (b) Performance on the KSC dataset. (c) Performance on the UP dataset. (d) Performance on the SA dataset.

with SVM and 3DCNN. Different datasets have different numbers of labeled pixels. Specifically, we progressively change the proportion of training data on the IP and KSC datasets from 1% to 5%, and progressively change the proportion of training data on the UP and SA datasets from 0.1% to 5%. The experimental results are demonstrated in Fig. 10. In the situation of extremely small training data, 3DCNN is prone to overfitting, seriously affecting the classification performance. Specifically, the classification results of 3DCNN are significantly superior to SVM on the IP, KSC, and UP datasets. But its superiority has not been clearly demonstrated on the SA dataset. It can be seen that our mutual guide method outperforms its counterparts overwhelmingly on the four datasets, suggesting the effectiveness and robustness of our mutual guide method for HIC with small training data.

We test the performance of the mutual guide with respect to the progressive change of the proportion threshold of the high confidence data. The results are given in Table V. The classification performance of the mutual guide method on the IP, KSC, and SA datasets achieves the best when the high confidence data threshold is set to 0.5%. The performance on the UP dataset achieves the best when the threshold is set to 0.7%. In order to maintain a consistent hyperparameter

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setting, we use 0.5% as the threshold indiscriminately for the four datasets.

We experimentally compare four mutual guide implementational approaches, i.e., the PCA and ELM combination, the KPCA and ELM combination, the LDA and ELM combination, and the proposed SuperPCA and ELM combination. The experimental results are given in Fig. 11. Though the first three combinations achieve acceptable performance, suggesting the effectiveness and robustness of our mutual guide framework, the SuperPCA and ELM combination outperforms them. This validates the effectiveness of SuperPCA as a feature extractor. This also reflects that different implementation approaches would result in different performances of the mutual guide framework, and implies that more comprehensive implementation approaches are supposed to render more effective results.

V. Conclusion

This article has developed a general yet effective mutual guide framework based on two base classifiers. The only requirement for the two base classifiers is that they have different random initializations, which allow them to learn different aspects from the same training data. The mutual guide framework alternates training and classification iteratively to improve the training and classification performance gradually, rendering an effective training strategy based on a small set of training data. Different from semi-supervised learning, the mutual guide framework has remarkable tolerance to intermediate noisy labels, leading to stronger robustness. This article has implemented the mutual guide framework based on SuperPCA and ELMs and forms a novel practical method suitable for the HIC with small training data. Experiments performed on the four public datasets have demonstrated that our mutual guide framework outperforms state-of-the-art classification methods. Last but not least, our mutual guide framework establishes a baseline for HIC with small training data, and more qualified base classifiers can be employed or developed for validation.

The proposed mutual guide is a generic framework. To implement the mutual guide framework, we have adopted a feature extractor and classifier combination scheme, which takes SuperPCA as the feature extractor and ELM as the classifier. Though achieving effective results, the SuperPCA and ELM combination is just one option for implementing the mutual guide framework. For one thing, alternative feature extractors can be used to replace the SuperPCA and other classifiers satisfying the Proposition 1 can be employed as substitutes for the ELMs. For another, end-to-end classification models - instead of the feature extractor and classifier combination - satisfying the Proposition 1 can be used to implement the mutual guide framework. Therefore, in our future work, we will investigate the principled strategy of adapting classification models for effectively implementing the mutual guide framework. Additionally, different from this study in which we use the same architecture for two base classifiers, we will investigate the possibility of characterizing the two base classifiers by different architectures, and thus further explore the representational power of the mutual guide framework.

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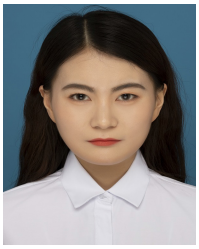
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